

09_functions

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1 Functions

1.1 Builtin Functions

```
[4]: prices = [100, 200, 300, 400, 650, 750]
     sum(prices)
```

[4]: 2400

```
[7]: prices = [100, 200, 300, 400, 650, 750, 850, 900, 950, 970, 985, 990, 995, 999]
     (len(prices))
```

[7]: 14

```
[9]: prices = [100, 200, 300, 400, 650, 750, 850, 900, 950, 970, 985, 990, 995,]
     max(prices)
```

[9]: 995

```
[12]: min(prices)
```

[12]: 100

```
[15]: mean = sum(prices)/len(prices)
     print(mean)
```

695.3846153846154

1.2 User defined functions

```
[24]: def great(name):
     """This function will greet the person with the name provided as an_
     ↪argument."""
     print("Hello, " + name + "! Assalamualaikum!")
```

```
[26]: great("Abdul Rahman")
```

Hello, Abdul Rahman! Assalamualaikum!

1.2.1 Defining Mean Function

```
[27]: def mean(numbers):  
        """This function will calculate the mean of the numbers provided as an  
        ↪argument."""  
        return sum(numbers)/len(numbers)
```

```
[29]: prices = [100, 200, 300, 400, 650, 750, 850, 900, 985, 990, 995,]  
mean(prices)
```

```
[29]: 647.2727272727273
```

```
[30]: def mean(numbers):  
        """This function will calculate the mean of the numbers provided as an  
        ↪argument."""  
        return round(sum(numbers)/len(numbers), 2)
```

```
[31]: prices = [100, 200, 300, 400, 650, 750, 850, 900]  
mean(prices)
```

```
[31]: 518.75
```

```
[36]: def add(a, b, c):  
        """This function will add three numbers provided as arguments."""  
        return a + b + c
```

```
[37]: add(256, 512, 1024)
```

```
[37]: 1792
```

```
[38]: def introduce(name, age, city):  
        """This function will introduce a person with the name, age, and city  
        ↪provided as arguments."""  
        print("Hello, My name is " + name + ". I am " + str(age) + " years old and  
        ↪I live in " + city + ".")
```

```
[39]: introduce("Abdul Rahman", 20, "Lahore")
```

Hello, My name is Abdul Rahman. I am 20 years old and I live in Lahore.

1.2.2 Function without any parameter

```
[40]: def say_hello():  
        """This function will print a greeting message."""  
        print("Hello, Abdul Rahman! Welcome to Python programming.")
```

```
[41]: say_hello()
```

Hello, Abdul Rahman! Welcome to Python programming.

```
[42]: def say_hi():  
        """This function will print a greeting message."""  
        print("Hi, Abdul Rahman! Welcome to Codanics Python ka Chilla course.")
```

```
[43]: say_hi()
```

Hi, Abdul Rahman! Welcome to Codanics Python ka Chilla course.

```
[44]: def great(name="Abdul Rahman"):  
        """This function will greet the person with the name provided as an  
        ↪argument."""  
        print("Assalamualaikum, " + name + "! Welcome to Python programming.")
```

```
[46]: great("Abdul Rahman")
```

Assalamualaikum, Abdul Rahman! Welcome to Python programming.

1.2.3 For Addition

```
[48]: def add(a, b):  
        """This function will add two numbers provided as arguments."""  
        return a + b  
add(124, 236)
```

```
[48]: 360
```

```
[51]: def subtract(a, b):  
        """This function will subtract two numbers provided as arguments."""  
        return a - b  
subtract(986, 248)
```

```
[51]: 738
```

```
[54]: def multiply(a, b):  
        """This function will multiply two numbers provided as arguments."""  
        return a * b  
multiply(568, 1498)
```

```
[54]: 850864
```

```
[56]: def divide(a, b):  
        """This function will divide two numbers provided as arguments."""  
        return a / b  
divide(457, 23)
```

```
[56]: 19.869565217391305
```

```
[ ]:
```